Group Topic: **Big Data Analysis for Rainfall-Based Flood Risk Prediction in Albay using SYNOP Data**

Week 4

Git Repo/ Colab Link: https://github.com/PotatoKevz/Big-Data-Analysis-Repository

Date: May 15, 2026

Objectives :

The main goal of this phase is to enhance the flood risk prediction by:

- Developing a modular machine learning pipeline that integrates with external tools for better functionality and usability.
- Creating professional and interactive visualizations of the processed data and model results.
- Presenting clear algorithm performance metrics and visualizations to demonstrate project progress.

Introduction :

This laboratory activity serves to further enhance the practical skills of the students pertaining to data handling through developing a modular machine learning pipeline that is integrated with external tools, further enhancing the prior skills gained through recent laboratory practices of handling big data. This activity implements a Flood Risk Prediction System that analyzes SYNOP weather station data from multiple stations surrounding Albay. The system utilizes machine learning models to analyze rainfall patterns and predict the probability of flood risk events. The system integrates several technologies including Python-based data processing libraries, machine learning algorithms, and an interactive visualization dashboard. The result is a comprehensive platform that transforms raw weather data into meaningful insights that can support disaster preparedness and decision-making.

Methodology :

The activity catered to a data-driven classification approach to process data using historical SYNOP meteorological data, a basis for the project objective, which is to predict rainfall-based flood risk. The methodology was executed through a structured, end-to-end pipeline, and has also implemented changes from the previous activity's methods. Due to these, some approaches from this activity were slightly modified from the earlier task documentations. This week's activity involved primary steps that were done to accomplish; developing a pipeline that utilizes external tools, and visualizing processed data in a professional format.

1. Pipeline Development with External Tool Integration

- A core backend pipeline (flood_risk_pipeline.py) was developed to handle data loading, preprocessing, feature engineering, model training, and prediction generation.
- The pipeline outputs (predictions, metrics, and model artifacts) were designed to be easily consumed by external applications.
- Streamlit was integrated as the primary external tool to build an interactive web-based dashboard (dashboard.py).
- Additional external libraries integrated include:
    - **Plotly** – for creating interactive and dynamic visualizations.
    - **Joblib** – for saving and loading the trained machine learning model (rf_flood_model.pkl).
    - **Pandas** – for efficient data loading and manipulation between pipeline and dashboard.
- This integration established a complete workflow:

  SYNOP Data → Pipeline Processing → Model Prediction → Streamlit Dashboard.

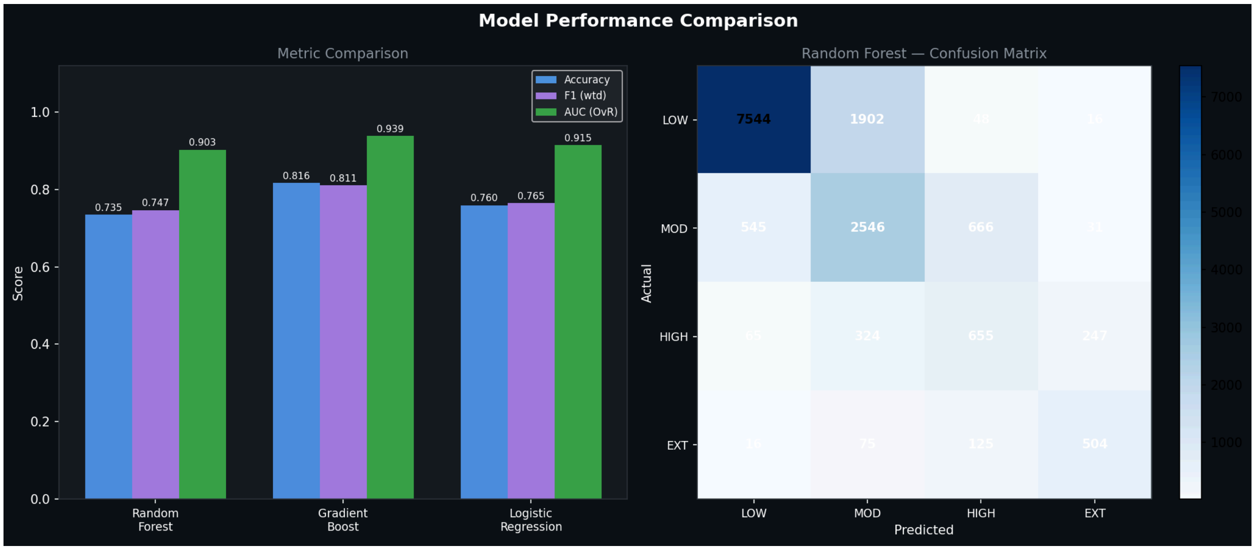
2. Professional Data Visualization

- All visualizations were generated using **Plotly** within the Streamlit dashboard to ensure high interactivity and professional quality.
- A clean, multi-tab dashboard layout was implemented with three main sections:
    - **Overview Tab**: Time series plots, risk class pie charts, rainfall histograms, and KPI cards.
    - **Model Performance Tab**: Comparative bar charts for algorithm metrics and feature importance visualizations.
    - **Extreme Events Tab**: Scatter plots highlighting high-risk and extreme rainfall occurrences.
- Interactive features such as date range filters, risk class selectors, hover tooltips, zoom/pan capabilities, and conditional formatting were added to enhance usability.
- A professional dark theme with consistent, accessible colors was applied throughout the dashboard.

Results and Analysis :

The system evaluated three machine learning models for flood and risk prediction. The results showed that the Gradient Boosting model achieved the best overall performance across multiple evaluation metrics.

Model	Accuracy	Weighted F1	AUC
Random Forest	73.48%	0.7469	0.903
Gradient Boost	81.64%	0.8107	0.939
Logistic Regression	75.96%	0.7654	0.915



The Gradient Boosting model achieved the highest accuracy of 81.64% and the highest AUC score of 0.939, indicating strong predictive capability and a good balance between precision and recall across flood risk categories.

This model was therefore selected as the best-performing algorithm from the flood prediction system.

Feature Importance Analysis

It revealed that rainfall-related features played the most significant role in predicting flood risk. The most influential predictors included:

- Rainfall lag features from Albay station
- 24-hour rainfall accumulation
- Rainfall data from nearby stations such as Catanduanes

These results indicate that recent rainfall patterns and upstream precipitation are key indicators of flood risk.

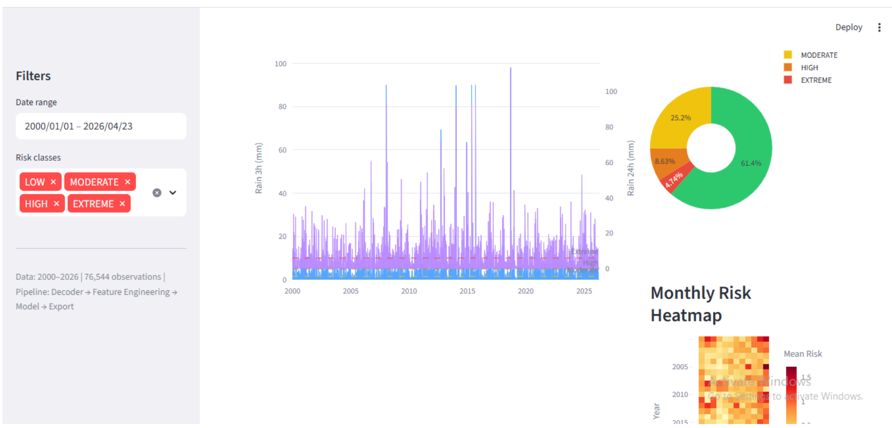
VISUALIZATION RESULTS

The system produced both interactive visualizations and static charts for analysis and reporting.

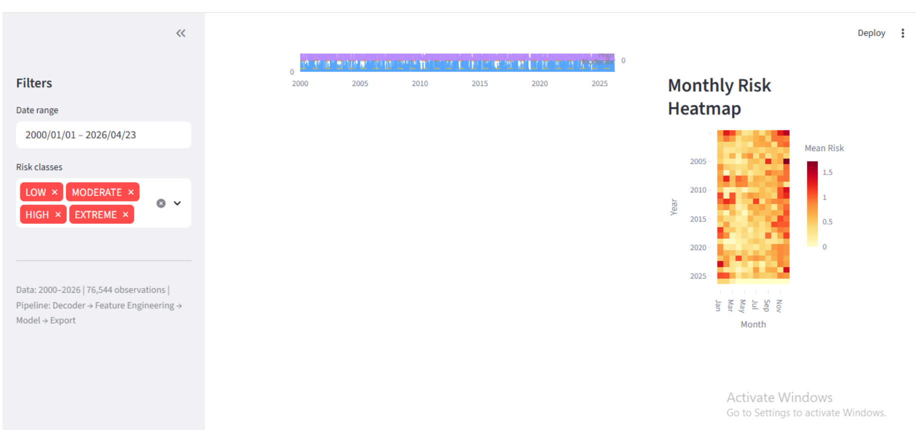
Interactive Dashboard Visualizations

Overview Tab

- Rainfall timeline showing precipitation levels over time
- Flood risk distribution pie chart

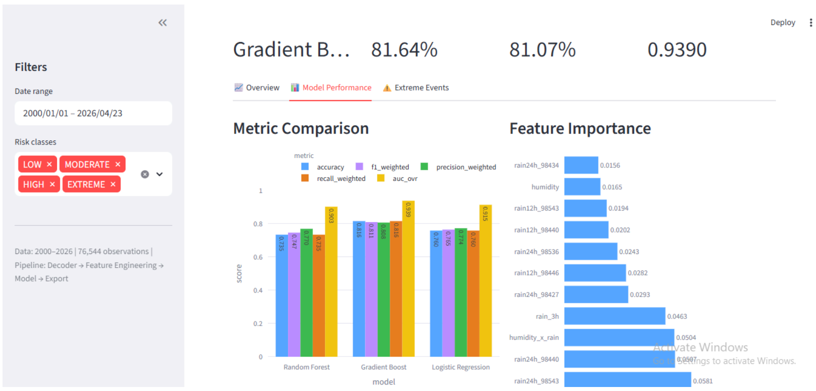


- Monthly risk heatmap
- Key performance indicator (KPI) cards showing high and extreme risk events

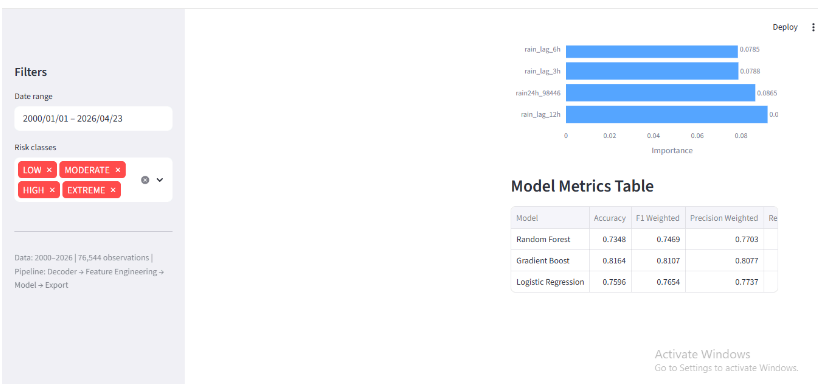


Model Performance Tab

- Grouped bar chart comparing performance of all models
- Feature importance charts displaying top predictive variables

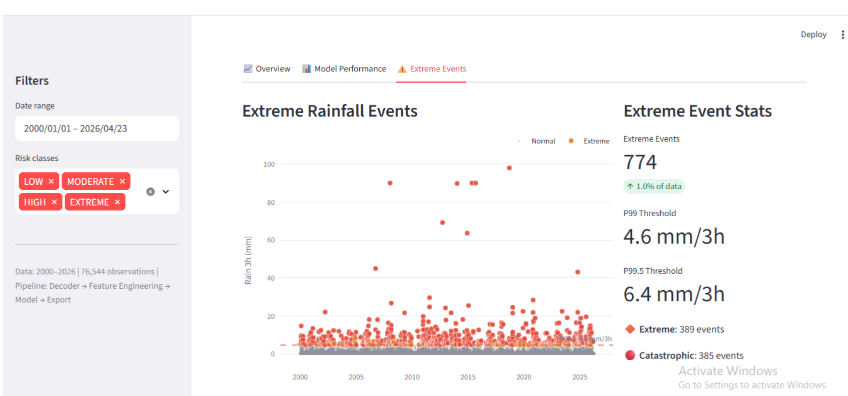


- Sortable metrics table for deeper analysis

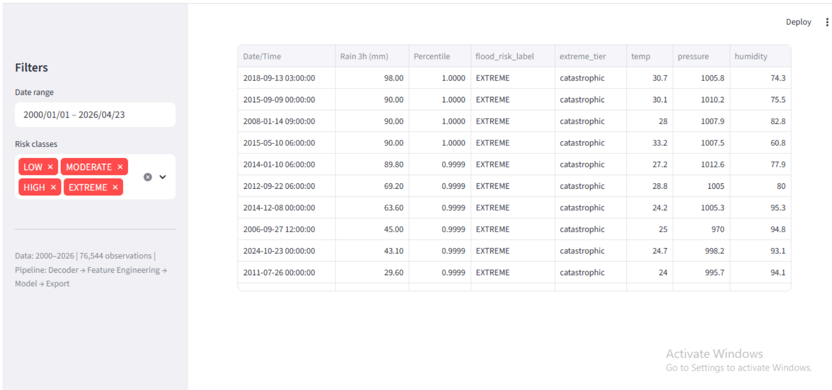


Extreme Events Tab

- Scatter plot highlighting extreme rainfall events
- Statistical summaries of high-risk occurrences



- Data tables listing significant events

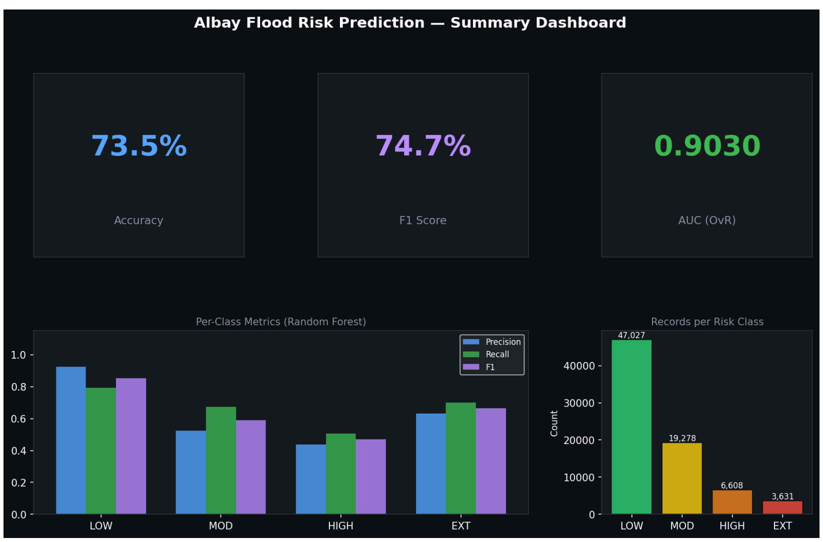


Date/Time	Rain 3h (mm)	Percentile	flood_risk_label	extreme_tier	temp	pressure	humidity
2018-09-13 03:00:00	98.00	1.0000	EXTREME	catastrophic	30.7	1005.8	74.3
2015-09-09 00:00:00	90.00	1.0000	EXTREME	catastrophic	30.1	1010.2	75.5
2008-01-14 09:00:00	90.00	1.0000	EXTREME	catastrophic	28	1007.9	82.8
2015-05-10 06:00:00	90.00	1.0000	EXTREME	catastrophic	33.2	1007.5	60.8
2014-01-10 06:00:00	89.80	0.9999	EXTREME	catastrophic	27.2	1012.6	77.9
2012-09-22 06:00:00	69.20	0.9999	EXTREME	catastrophic	28.8	1005	80
2014-12-08 00:00:00	63.60	0.9999	EXTREME	catastrophic	24.2	1005.3	95.3
2006-09-27 12:00:00	45.00	0.9999	EXTREME	catastrophic	25	970	94.8
2024-10-23 00:00:00	43.10	0.9999	EXTREME	catastrophic	24.7	998.2	93.1
2011-07-26 00:00:00	29.60	0.9999	EXTREME	catastrophic	24	995.7	94.1

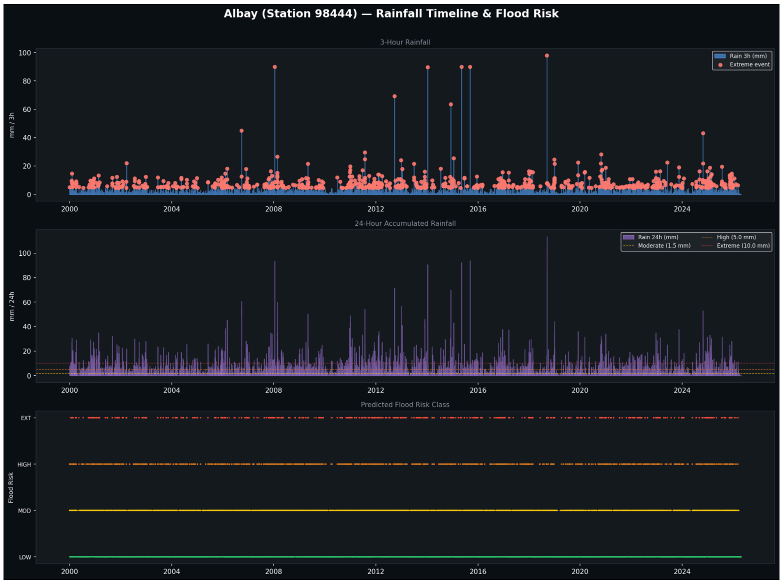
Static Output Visualization

The system also exports six static charts for documentation and reporting purposes :

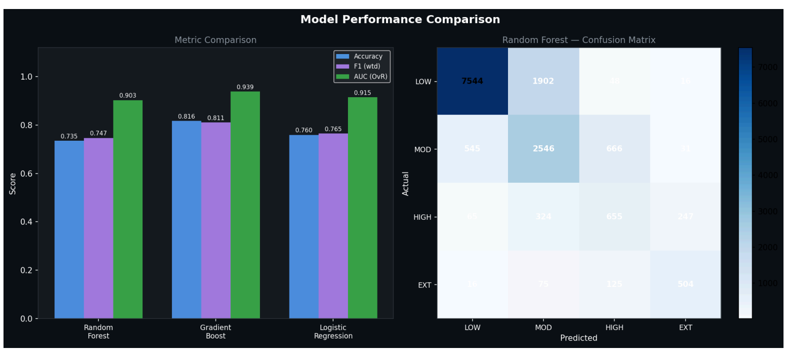
- Summary dashboard visualization



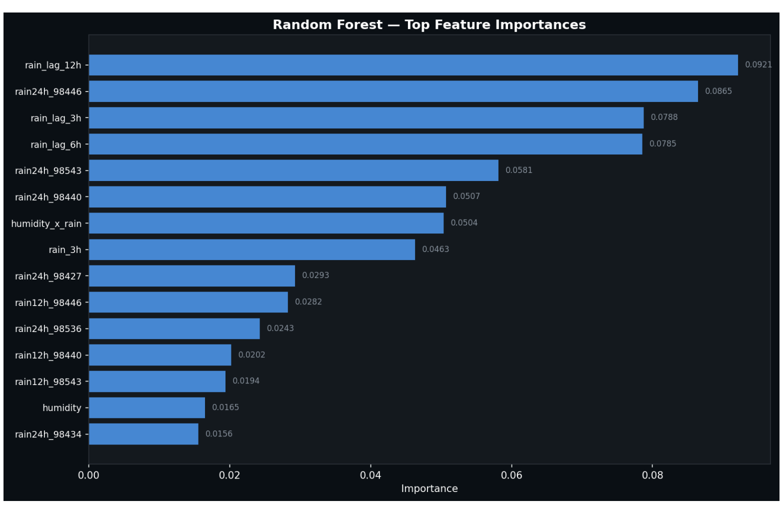
- Rainfall timeline chart



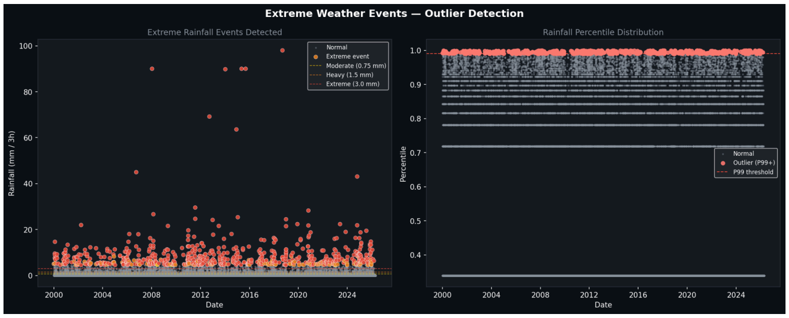
- Model comparison chart



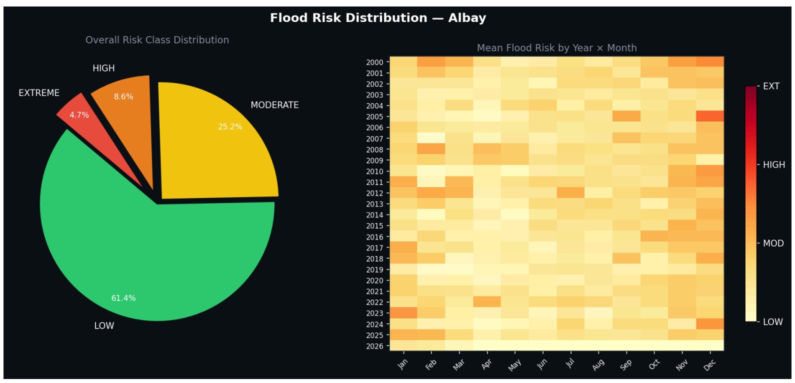
- Feature importance chart



- Extreme events scatter plot



- Risk distribution chart



These static charts provide a permanent record of the analysis and can be used in reports or presentations.

Challenges and Solutions :

During the implementation of the pipeline development and professional visualization tasks, the group encountered several technical and practical challenges. These challenges arose from the earlier methods and steps taken when accomplishing previous tasks. Below are the major ones identified along with the solutions applied:

Challenge 1: Target Leakage in Model Training

- Issue: Future rainfall features (e.g., rain_24h, rain_48h) were accidentally included in the feature set, causing unrealistically high model performance.
- Solution: Reviewed and removed all future-dependent features from the training data. Only past and current observations were used for prediction. Risk labels were generated strictly from the target 24-hour rainfall after feature engineering.

Challenge 2: Incorrect Rainfall Encoding (rrr field)

- Issue: Rainfall values in the SYNOP decoder were scaled 10x higher than actual values, leading to wrong risk classification and misleading visualizations.
- Solution: Fixed the rainfall decoding logic in synop_decoder.py and reprocessed the entire dataset. Risk thresholds were also adjusted accordingly to match DOST-PAGASA standards.

Challenge 3: Multi-Station Data Alignment

- Issue: Supporting stations had inconsistent timestamps and missing readings, making it difficult to merge data cleanly with the main Legazpi station (98444).
- Solution: Implemented 3-hour resampling (.resample('3h').mean()) and forward/backward filling with proper missing-data flags for each supporting station.

Challenge 4: Class Imbalance and Dashboard Bias

- Issue: The dataset was heavily dominated by LOW risk cases (~67%), causing models to bias toward the majority class and affecting visualization interpretability.
- Solution: Applied class_weight='balanced' in the pipeline and used stratified sampling. In the dashboard, added clear KPI cards and percentage breakdowns to highlight the imbalance.

Challenge 5: Dashboard Integration and Crashes

- Issue: The Streamlit dashboard crashed when loading outputs due to mismatched feature names between the pipeline and saved CSV files.
- Solution: Standardized output file formats, added error handling and data validation in dashboard.py, and ensured consistent column naming between flood_risk_pipeline.py and the dashboard.

Challenge 6: Creating Professional and Interactive Visualizations

- Issue: Basic Matplotlib charts were not interactive or polished enough for stakeholder presentation.
- Solution: Integrated Plotly within Streamlit to create zoomable, hover-enabled charts. Implemented a clean three-tab layout with filters, dark theme, and conditional formatting for a professional look.

Conclusion :

In conclusion, the Flood Risk Prediction System successfully demonstrated the use of machine learning and visualization techniques in analyzing rainfall data and predicting flood risks in Albay. By integrating predictive models, data processing, and an interactive dashboard, the system provided valuable insights into flood risk patterns.

Among the evaluated models, Gradient Boosting achieved the best performance, making it the most reliable predictor in the study. The visualization features also improved the accessibility and interpretation of analytical results.

Overall, the project highlights the importance of data-driven solutions in disaster preparedness and environmental monitoring. Future improvements may include adding more environmental variables, expanding the dataset, and enabling real-time flood monitoring.